Universitatea POLITEHNICA din București
Facultatea de Automatică și Calculatoare

Electronică analogică

Curs – Modulul 13

Stabilizatoare de tensiune

dr. ing. Răzvan Tătăroiu
razvan.tataroiu@upb.ro

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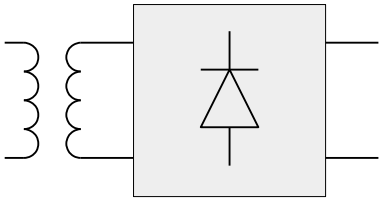
Sumar

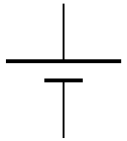
- “Surse de alimentare”
  - surse electronice de tensiune continuă
  - **redresor –> stabilizator** –> sarcină
    - baterie, altă sursă electronică

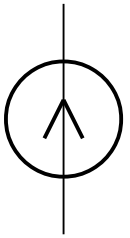
- Surse (stabilizatoare)
  - parametrice
  - lineare
  - în comutație

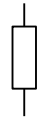
Schemă de principiu

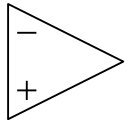


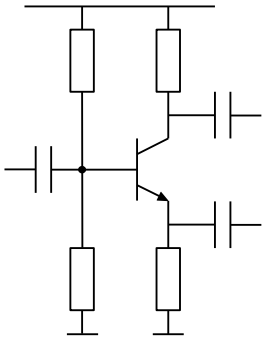






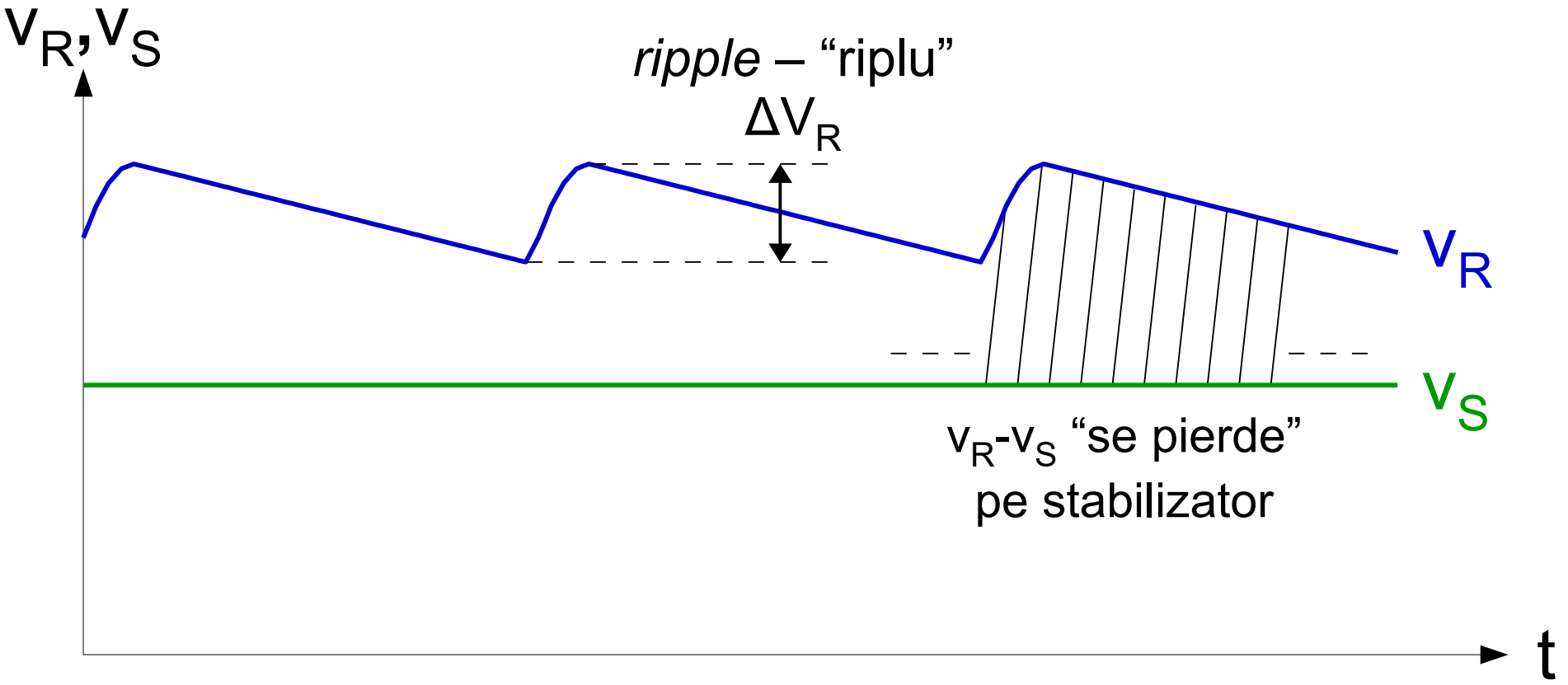






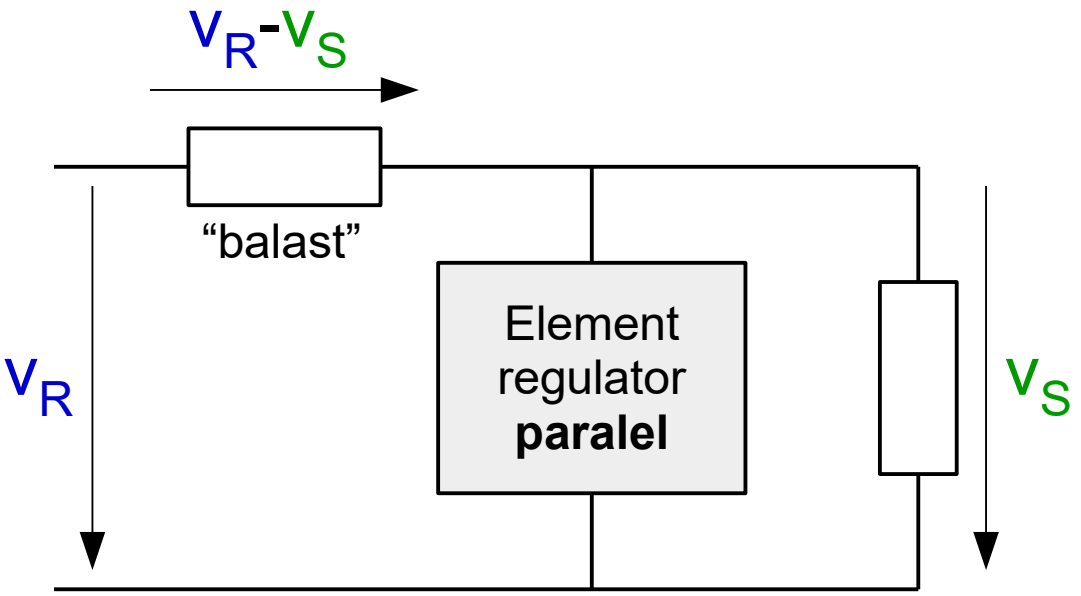
Principiu general (1)

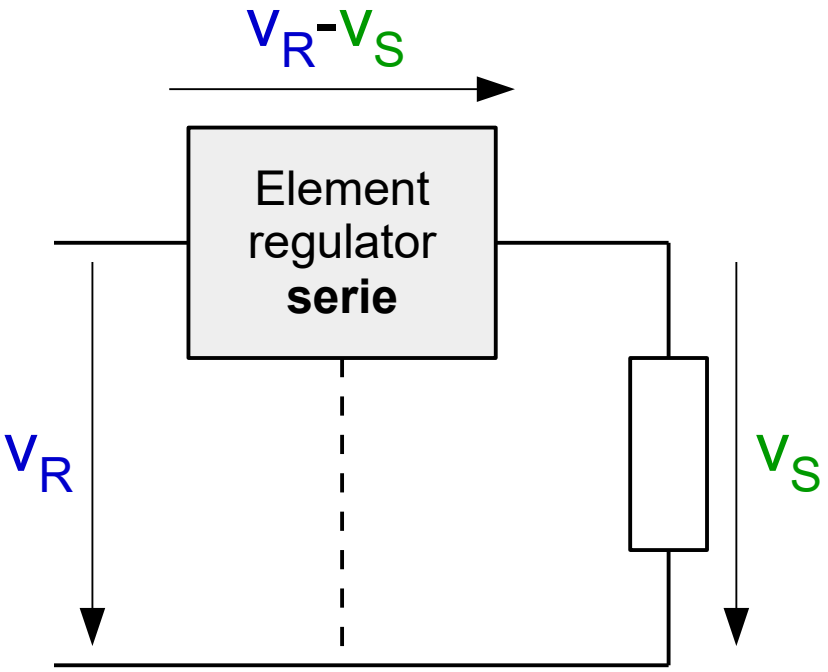




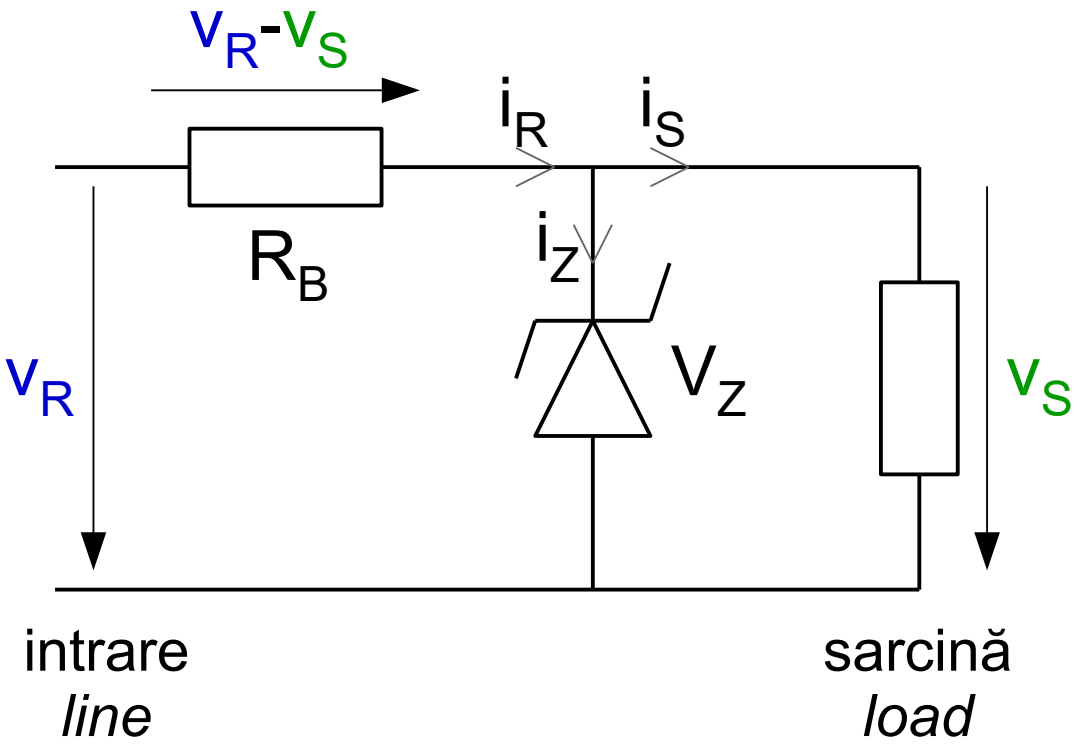
Principiu general (2)







Stabilizatoare parametrice (1)



parametrul:
tensiunea Zener

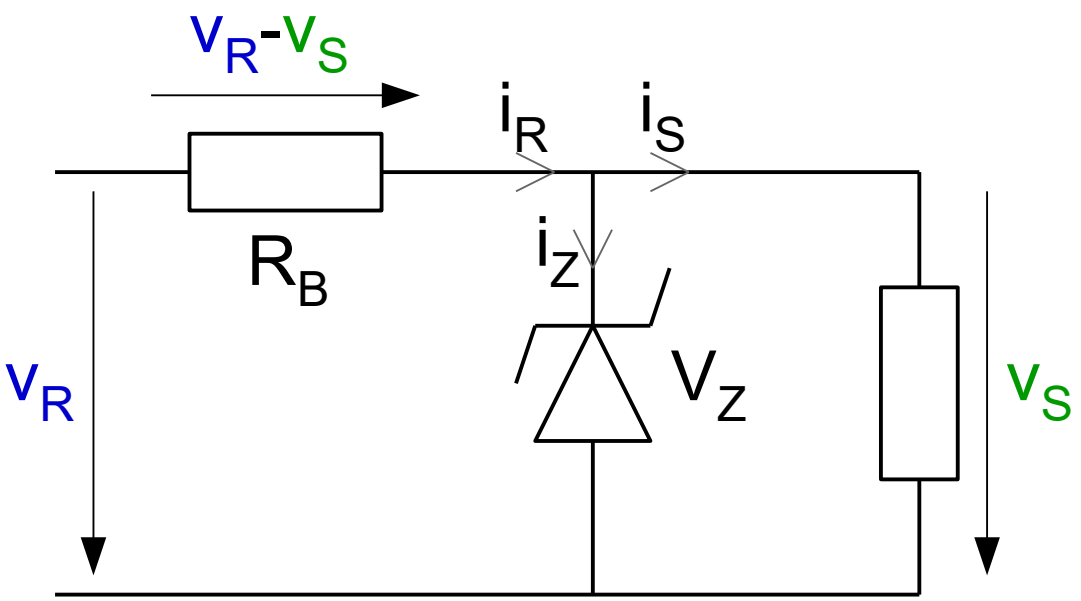
$$v_S = V_Z$$

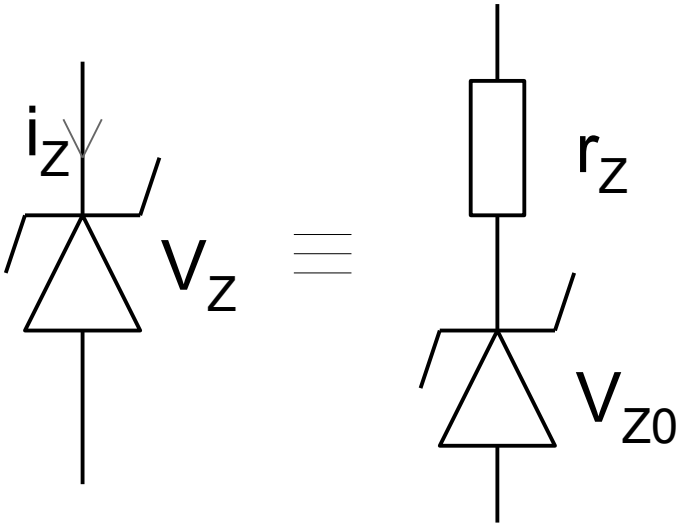
ideal: constantă
real:

$$V_Z = V_{Z0} + i_Z r_Z$$

OBS.:
Tensiunea stabilizată depinde atât de **tensiunea de intrare** cât și de **curentul** consumat **de sarcină**
=> indicatorii de performanță *Line Regulation, Load Regulation*
valabili pentru orice stabilizator

Stabilizatoare parametrice (2)

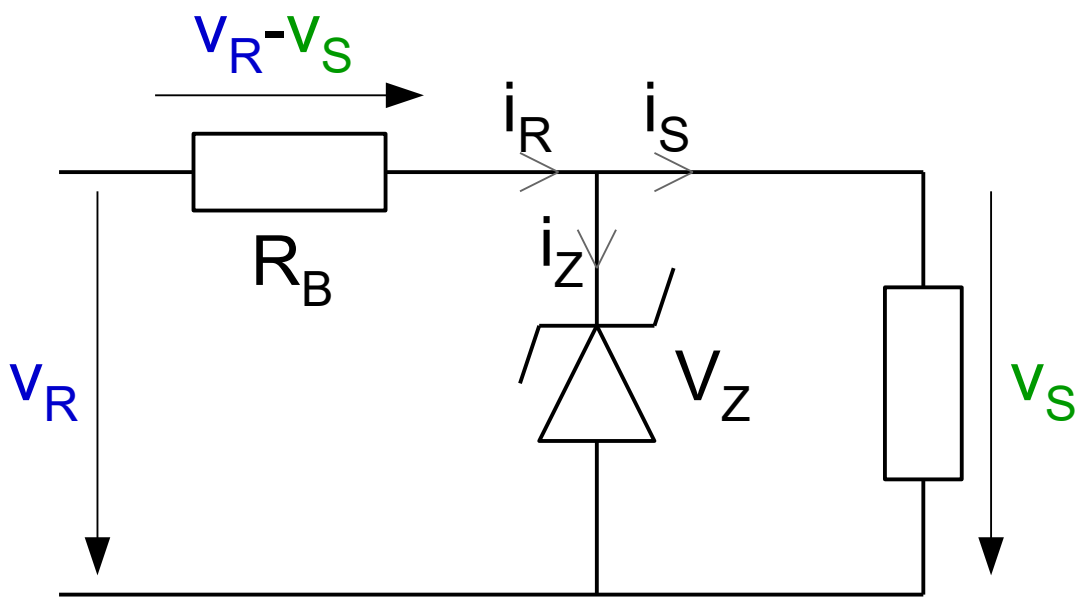




$$v_S = V_{Z0} + (i_R - i_S) r_Z$$

$$v_S = V_{Z0} + (v_R - v_S)\frac{r_Z}{R_B} - v_s \frac{r_Z}{R_S}$$

Stabilizatoare parametrice (3)



calculul rezistenței de balast:

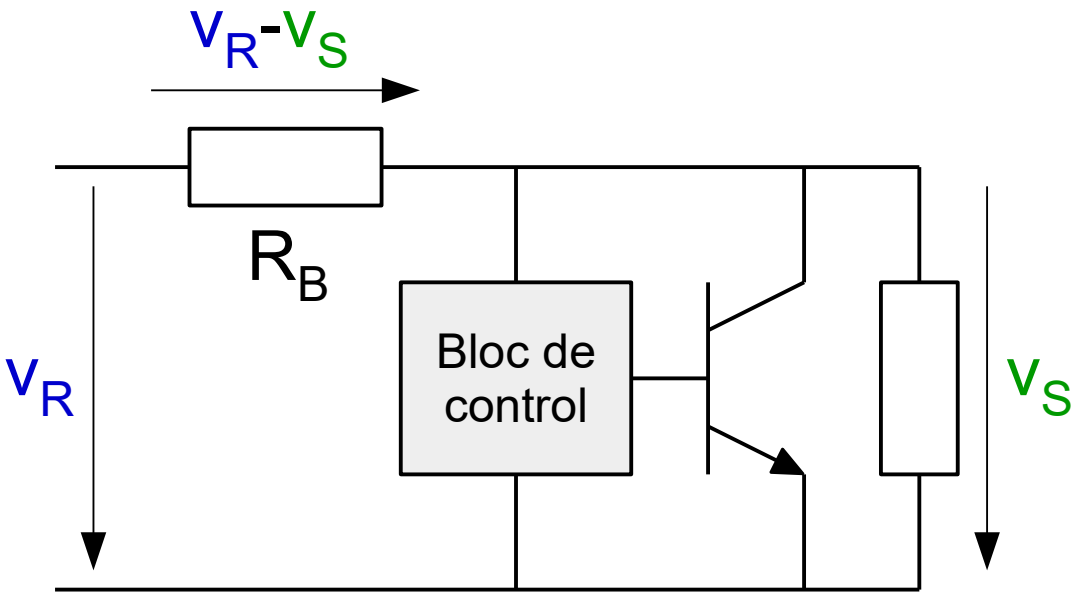
$$i_Z > I_{Zmin}$$

$$I_{Rmin} - I_{Smax} = I_{Zmin}$$

$$\frac{V_{Rmin} - V_Z}{R_B} = I_{Zmin} + I_{Smax}$$

dar la consum 0: $$I_{Zmax} = \frac{V_{Rmax} - V_Z}{R_B}$$ => randament foarte slab

Stabilizatoare-paralel (șunt)

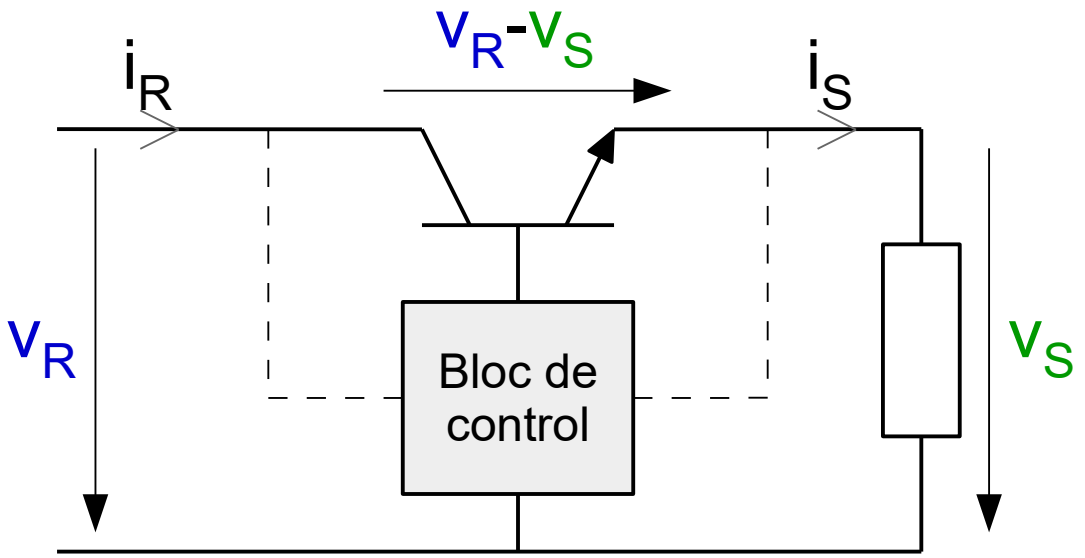


Emulează o diodă Zener cu performanțe superioare
- toleranță mică
- rezistență dinamică mică

Se utilizează doar în aplicații de **mică putere**

Cuvinte-cheie:
shunt voltage regulator
bandgap reference

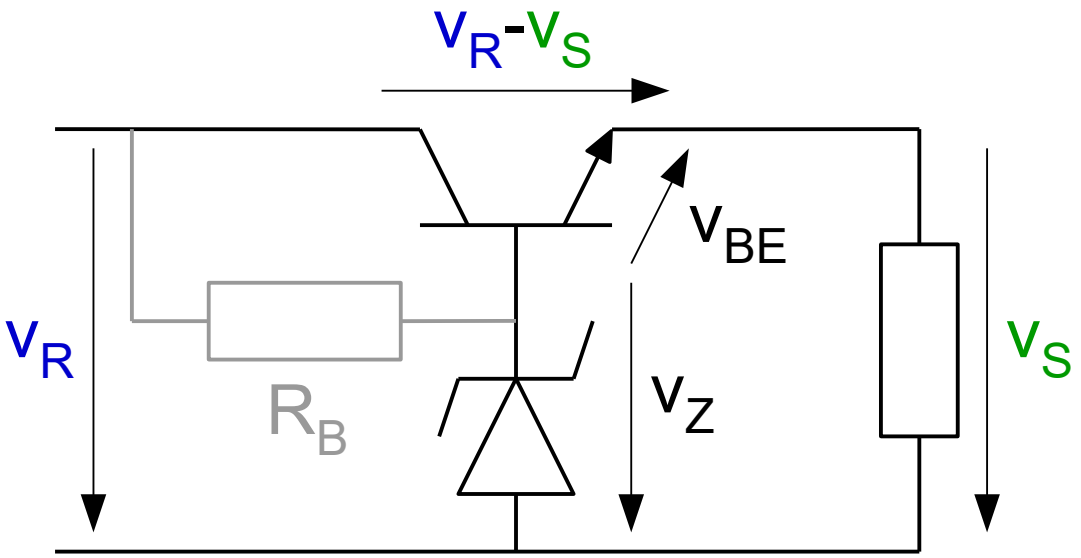
Stabilizatoare-serie



Avantaj: randament mai bun

curentul consumat din redresor ≈ curentul de sarcină

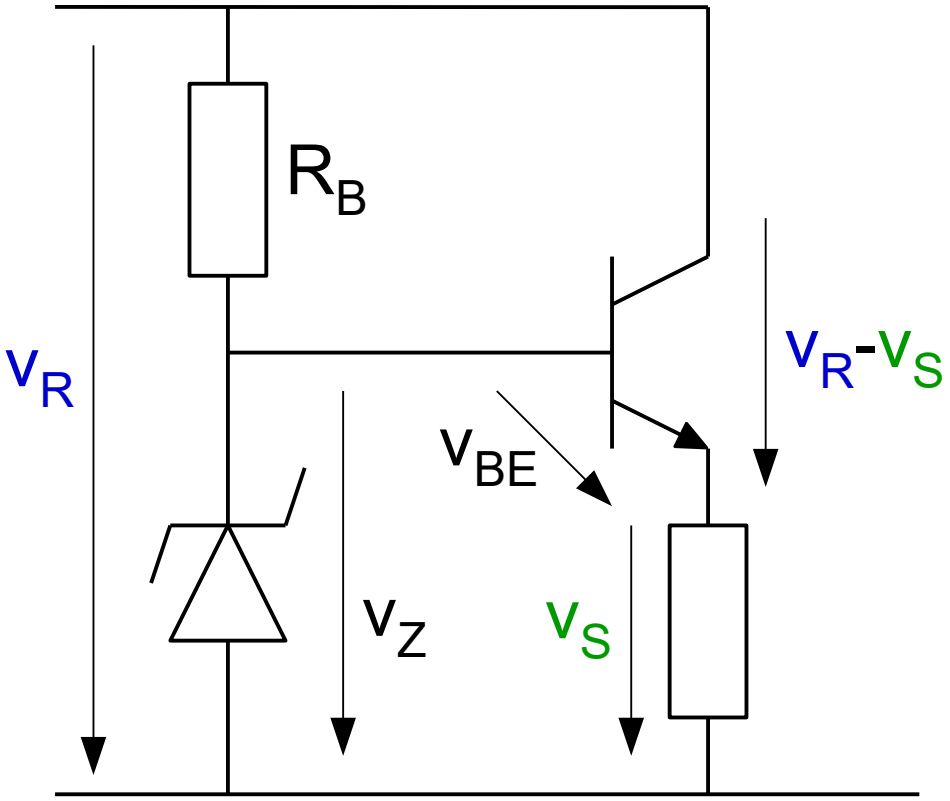
Zener + repetor pe emitor (1)



$$v_S = v_Z - v_{BE}$$

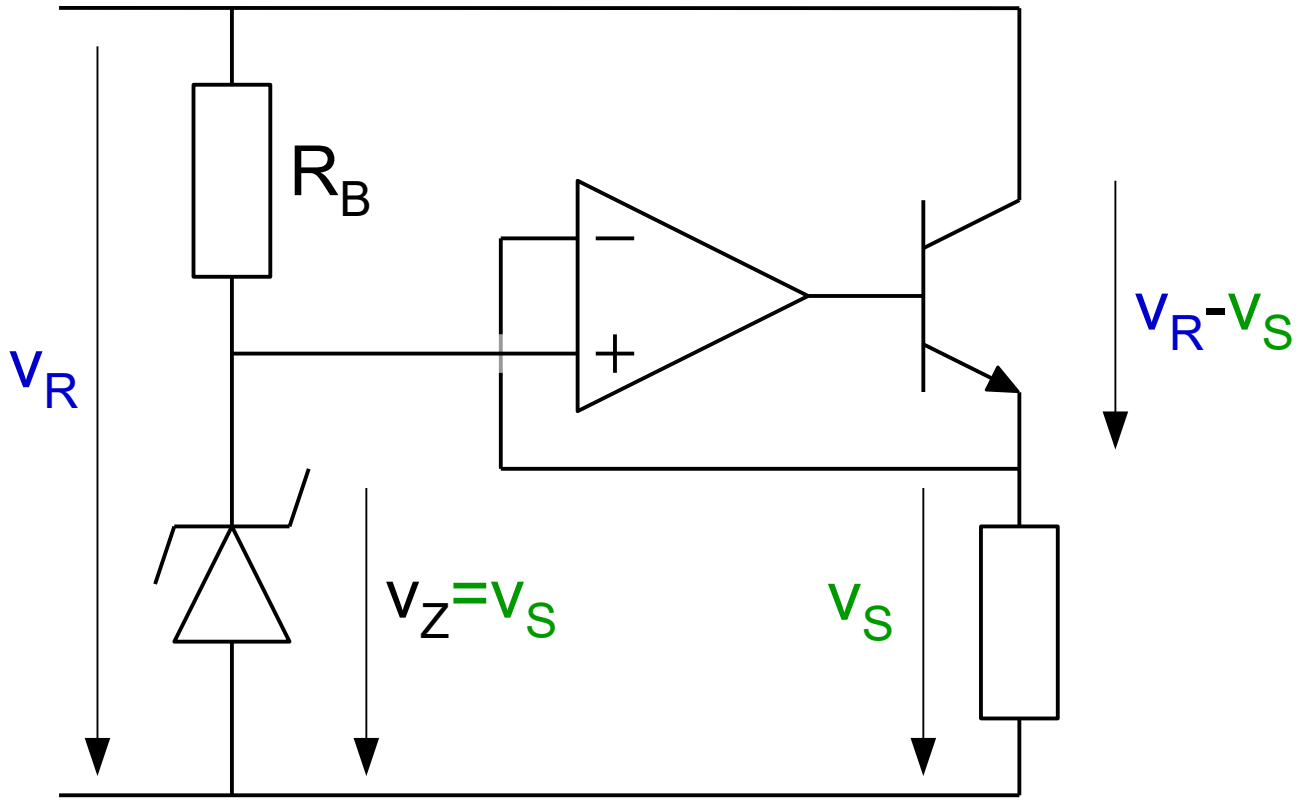
OBS.:
R_B de ~β ori mai mare decât
în cazul unui stabilizator-paralel

Zener + repetor pe emitor (2)



$$v_S = v_Z - v_{BE}$$

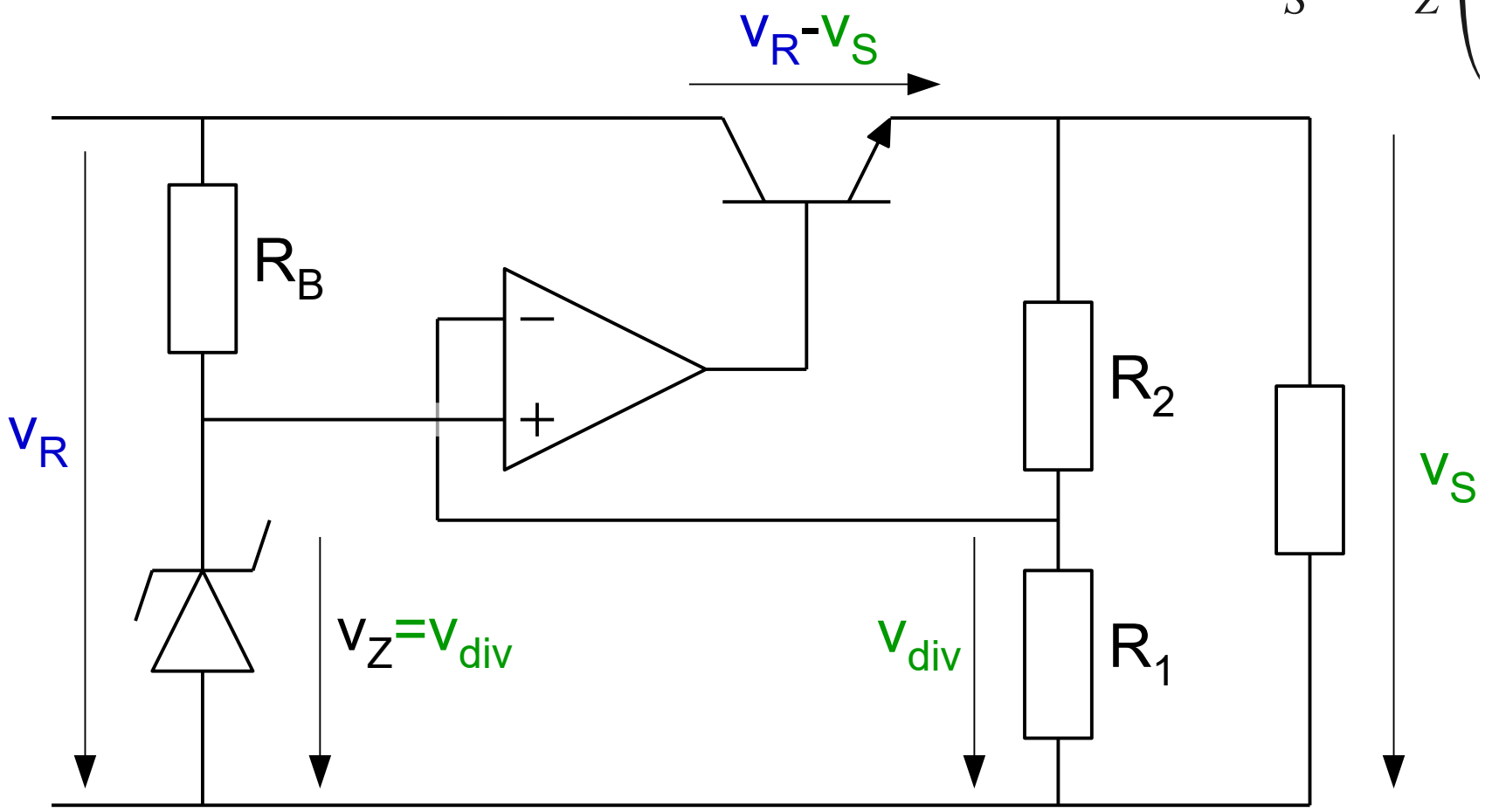
Zener + amplificator de eroare



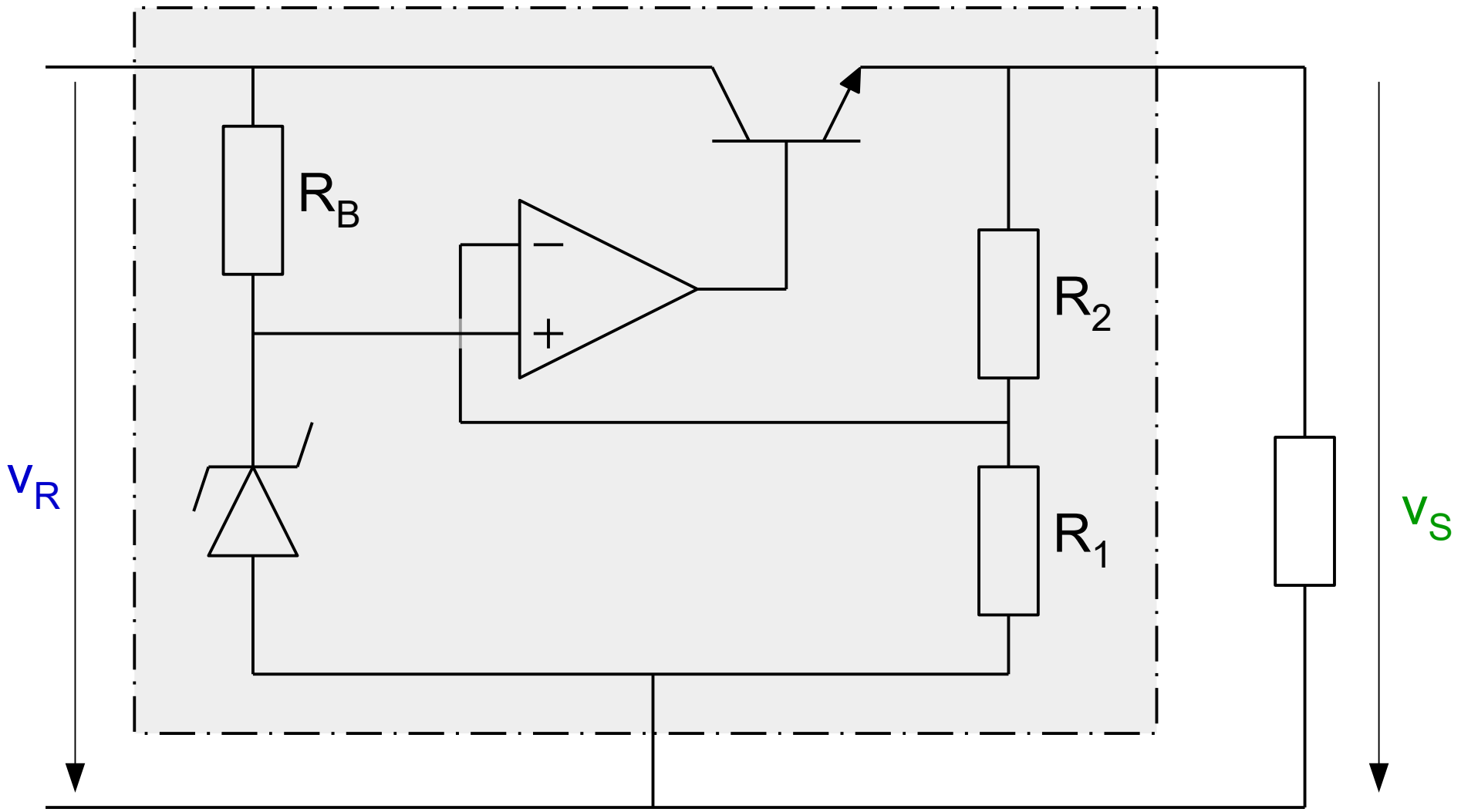
OBS.:
în loc de dioda Zener se poate utiliza o referință de tensiune, de ex. tip *bandgap*

Stabilizator cu referință fixă

$$v_S = v_Z\left(1+\frac{R_2}{R_1}\right)$$



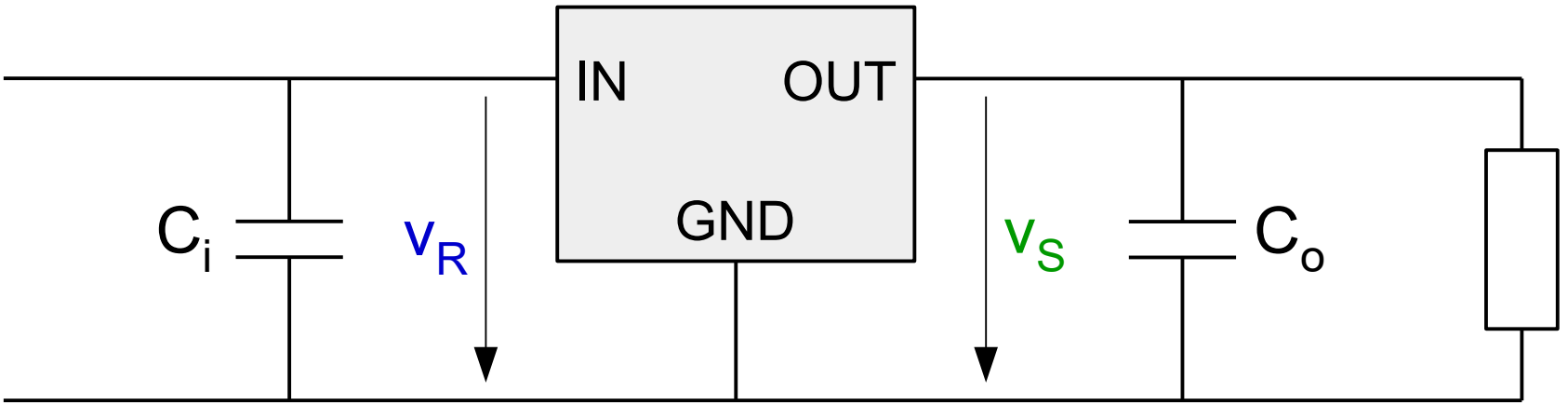
Stabilizator integrat fix (1)



Stabilizator integrat fix (2)

exemplu:
seria 78xx

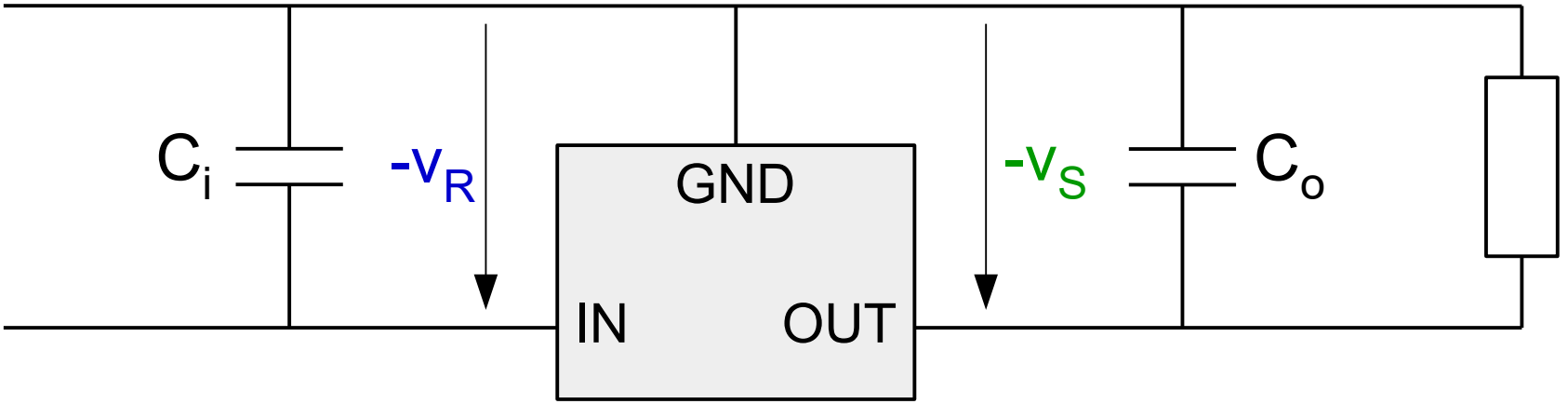
7805: 5 V
7809: 9 V
7812: 12 V
7815: 15 V
etc.



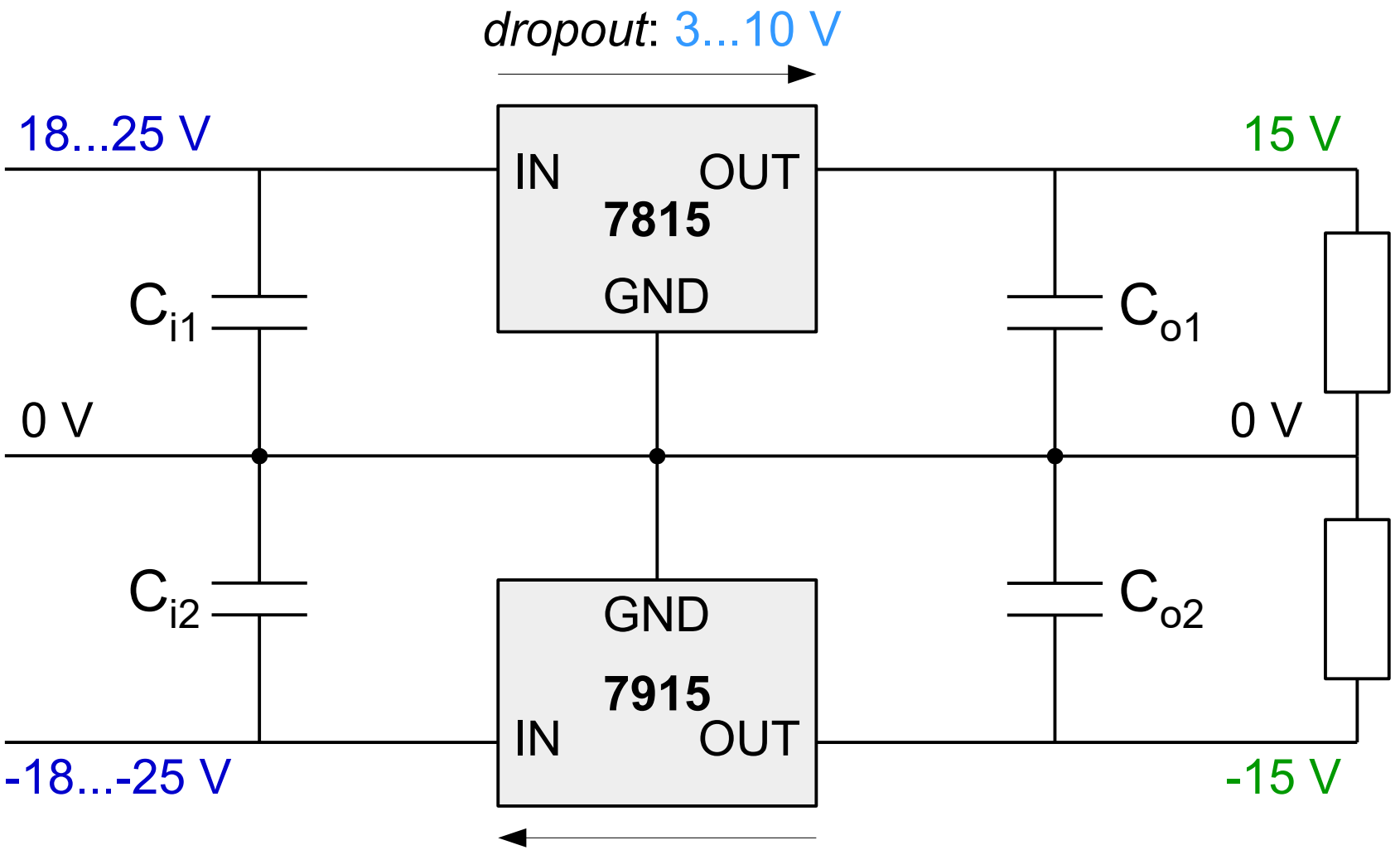
Stabilizator integrat fix negativ

exemplu:
seria 79xx

7905: -5 V
7909: -9 V
7912: -12 V
7915: -15 V
etc.



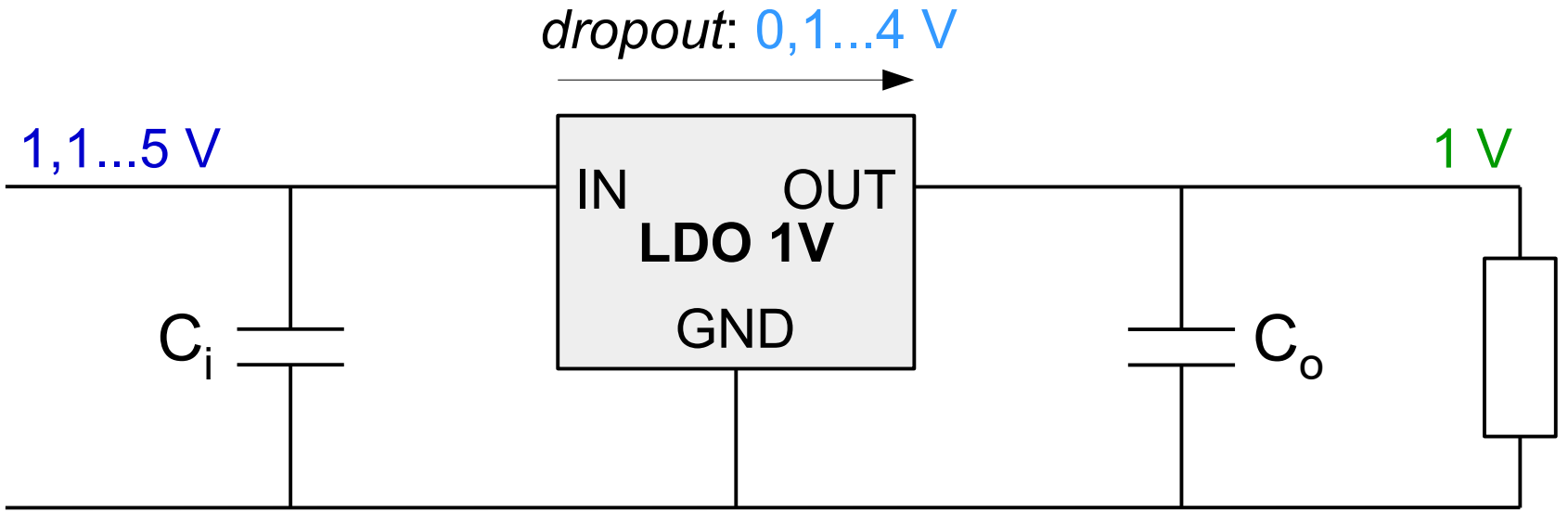
Alimentare simetrică



LDO (1)

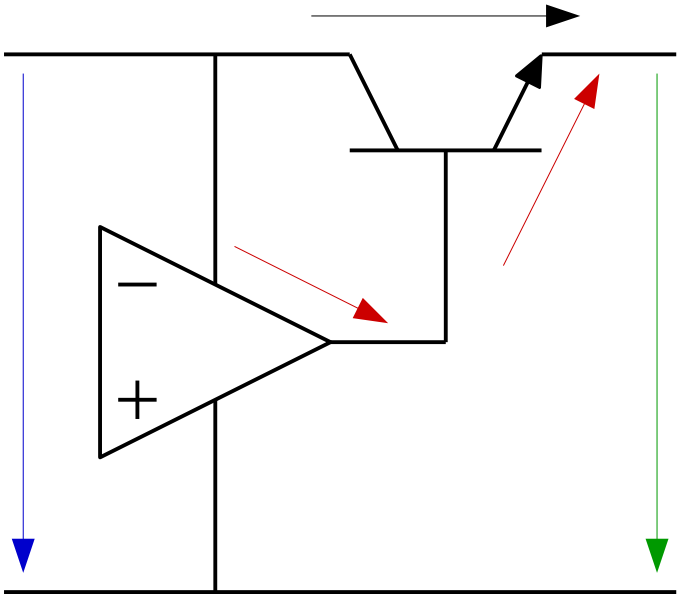
OBS.: În general stabilizatoarele necesită o tensiune de câțiva volți între intrare și ieșire pentru a putea asigura polarizarea schemei interne.

Pentru aplicații ce necesită tensiuni mici
se recomandă stabilizatoare tip LDO
(Low-DropOut Regulator)

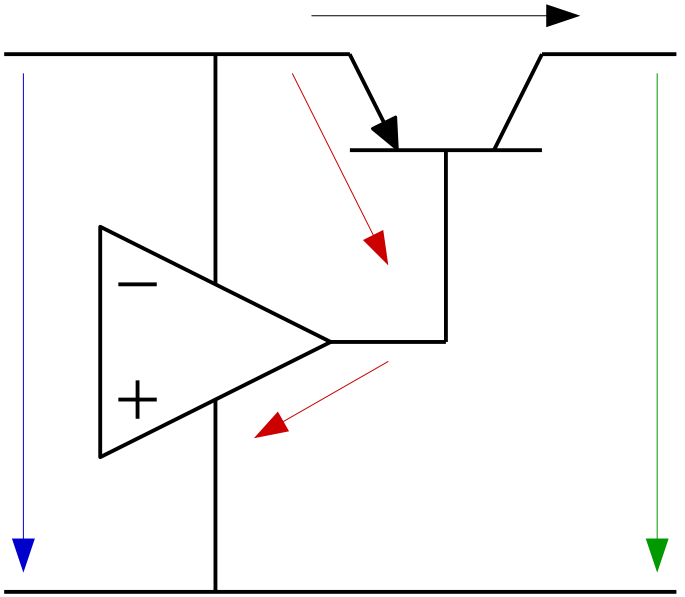


LDO (2)

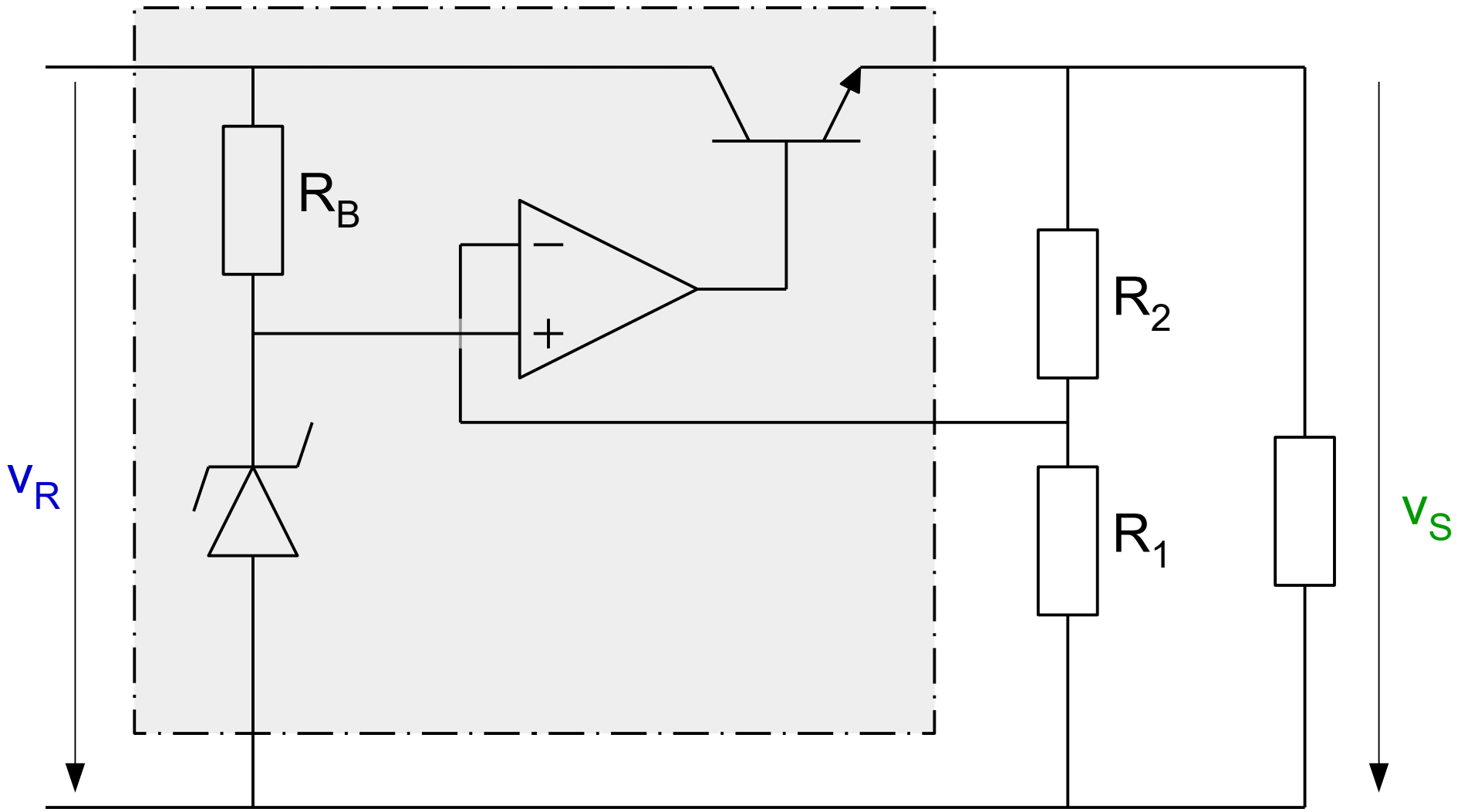
Stabilizator clasic (78xx)



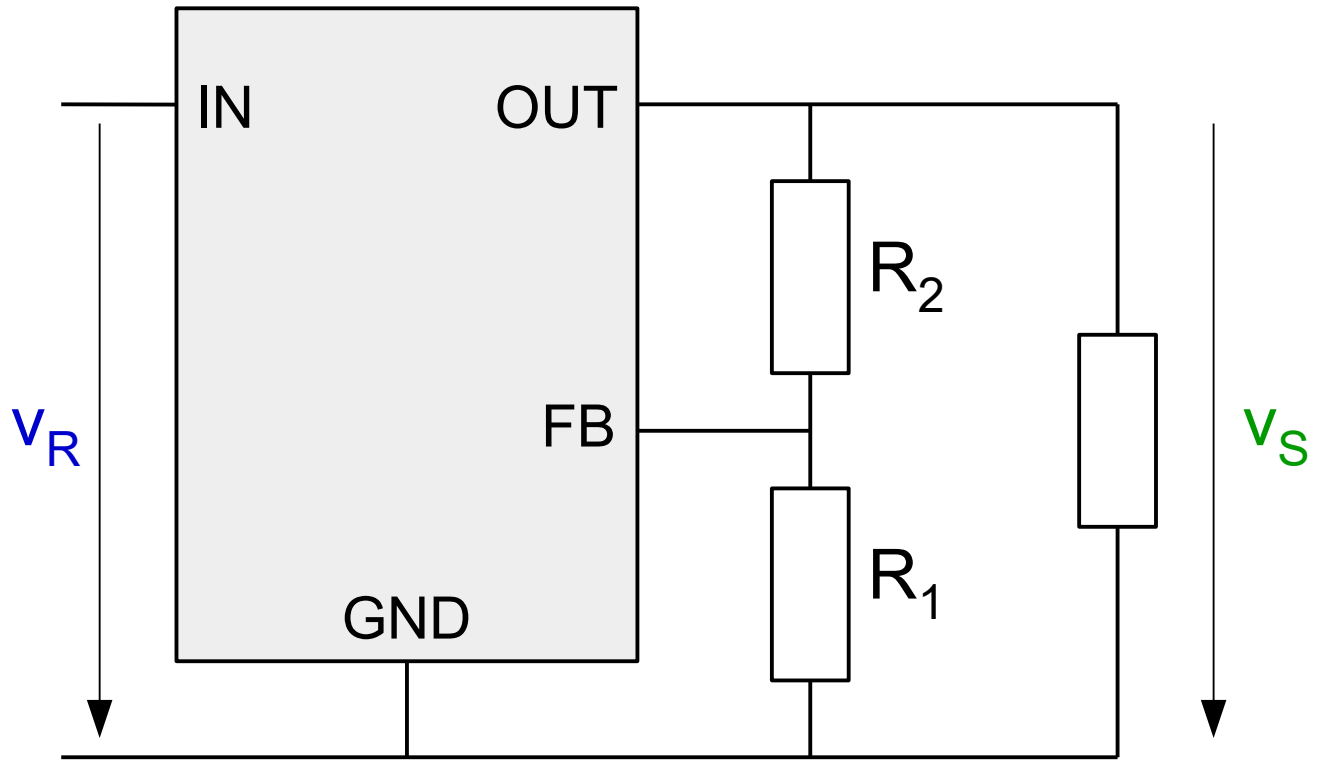
Low-DropOut Regulator



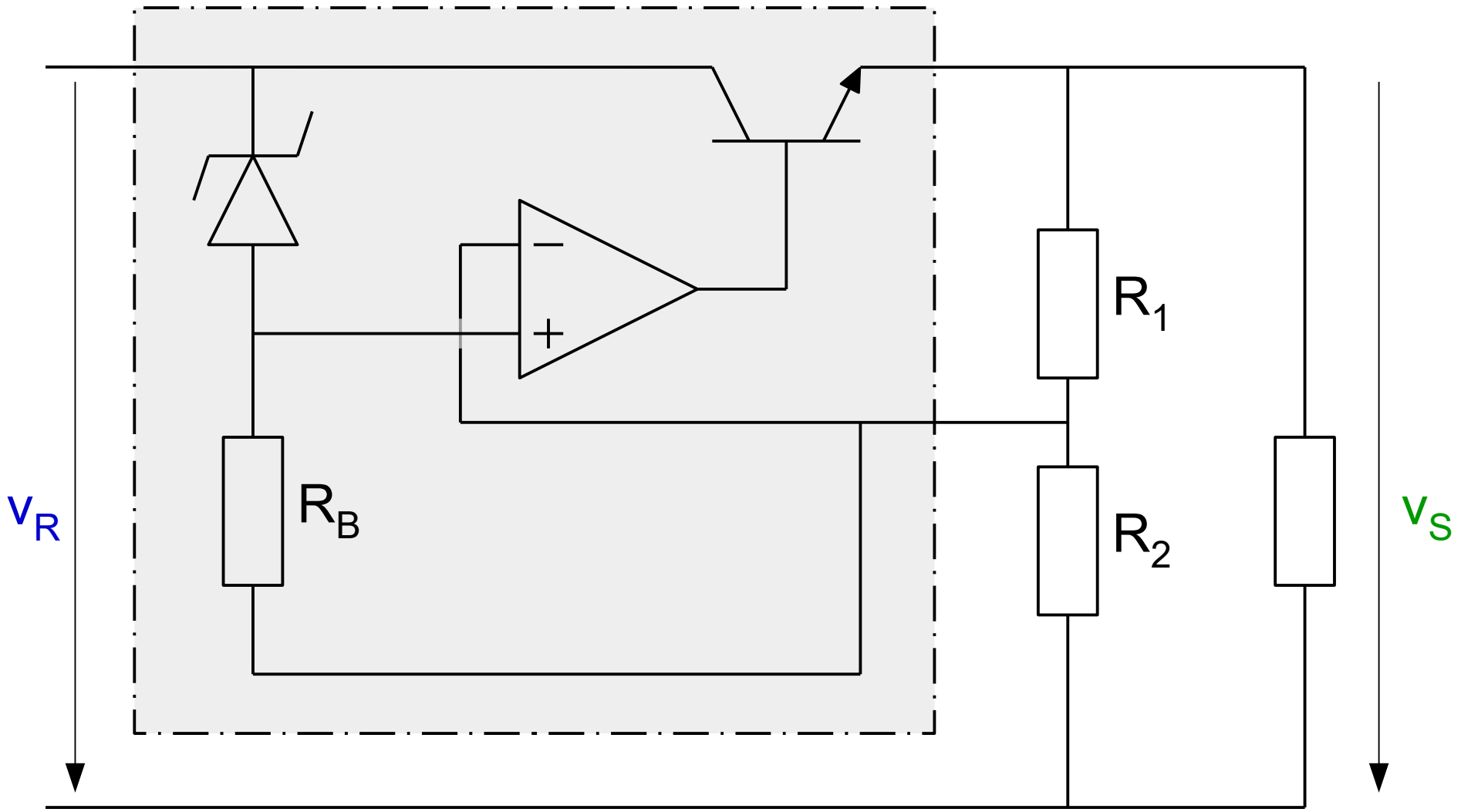
Stabilizator integrat ajustabil (1)



Stabilizator integrat ajustabil (2)

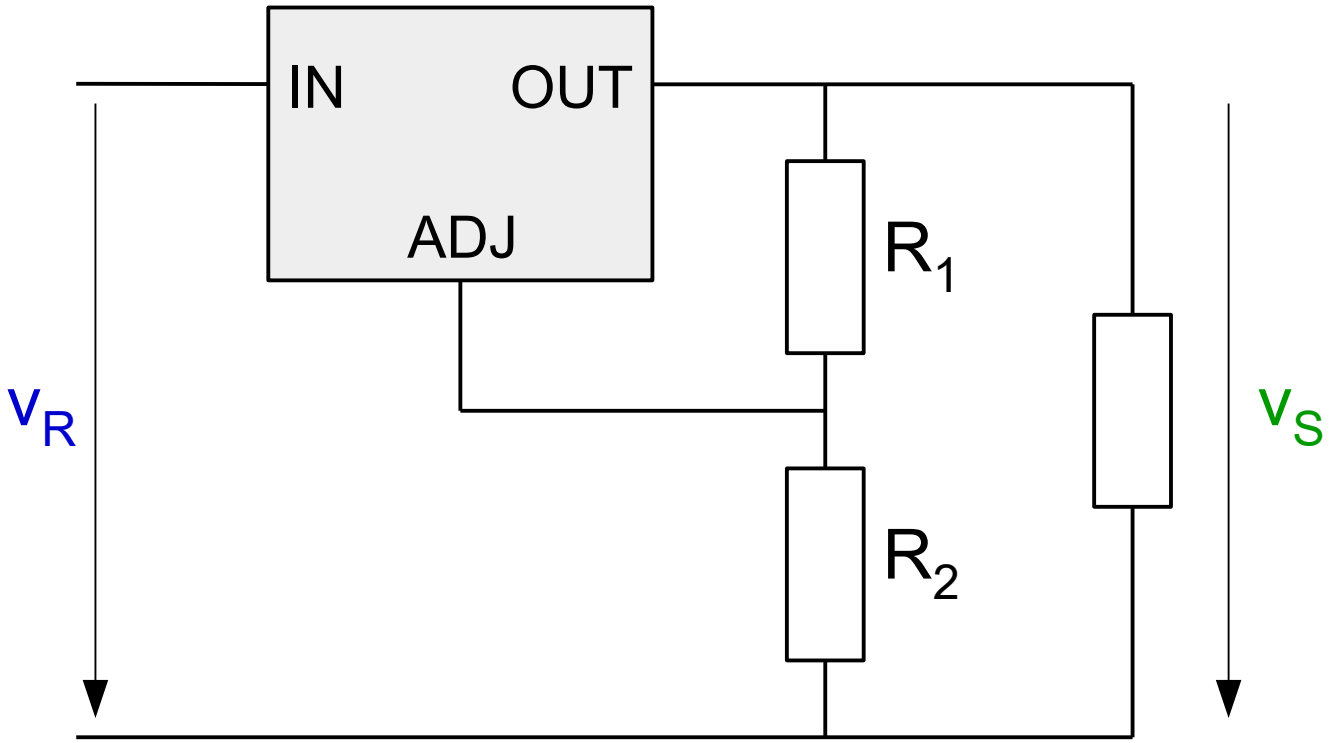


Stabilizator integrat ajustabil (3)

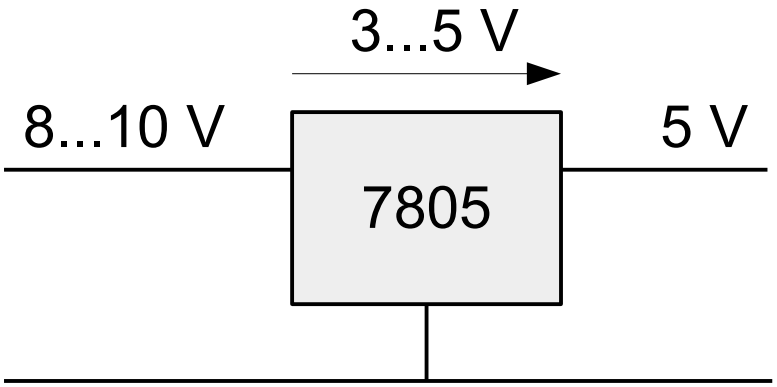


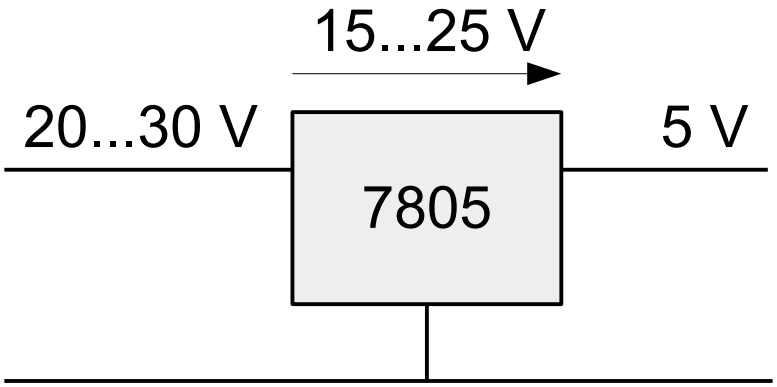
Stabilizator integrat ajustabil (4)

ex.:
LM317 (pozitiv)
LM337 (negativ)



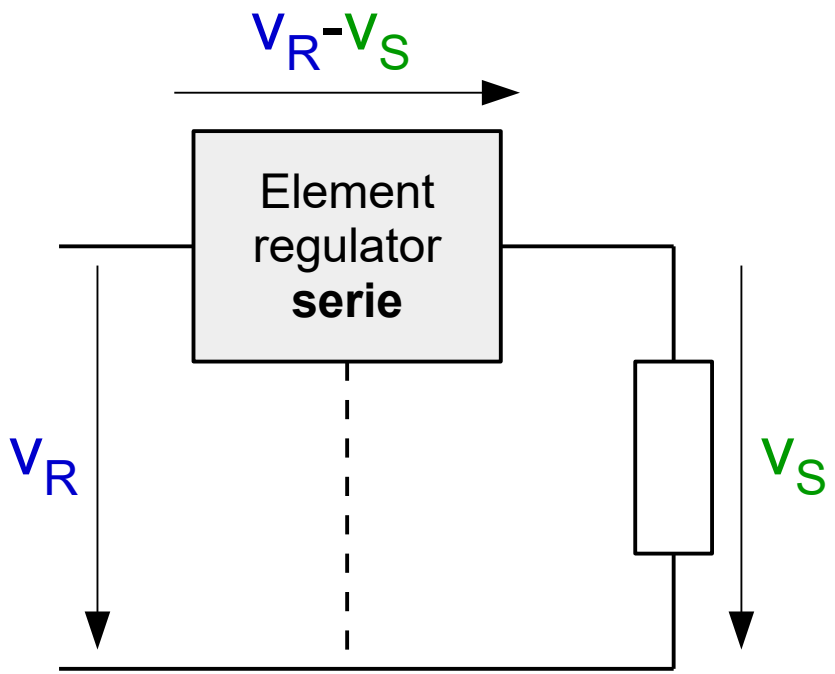
Considerații energetice





În general stabilizatoarele lineare sunt **ineficiente**.

Stabilizator în comutație (1)



Pe elementul regulator avem cădere de tensiune

$$v_{ERS} = v_R - v_S$$

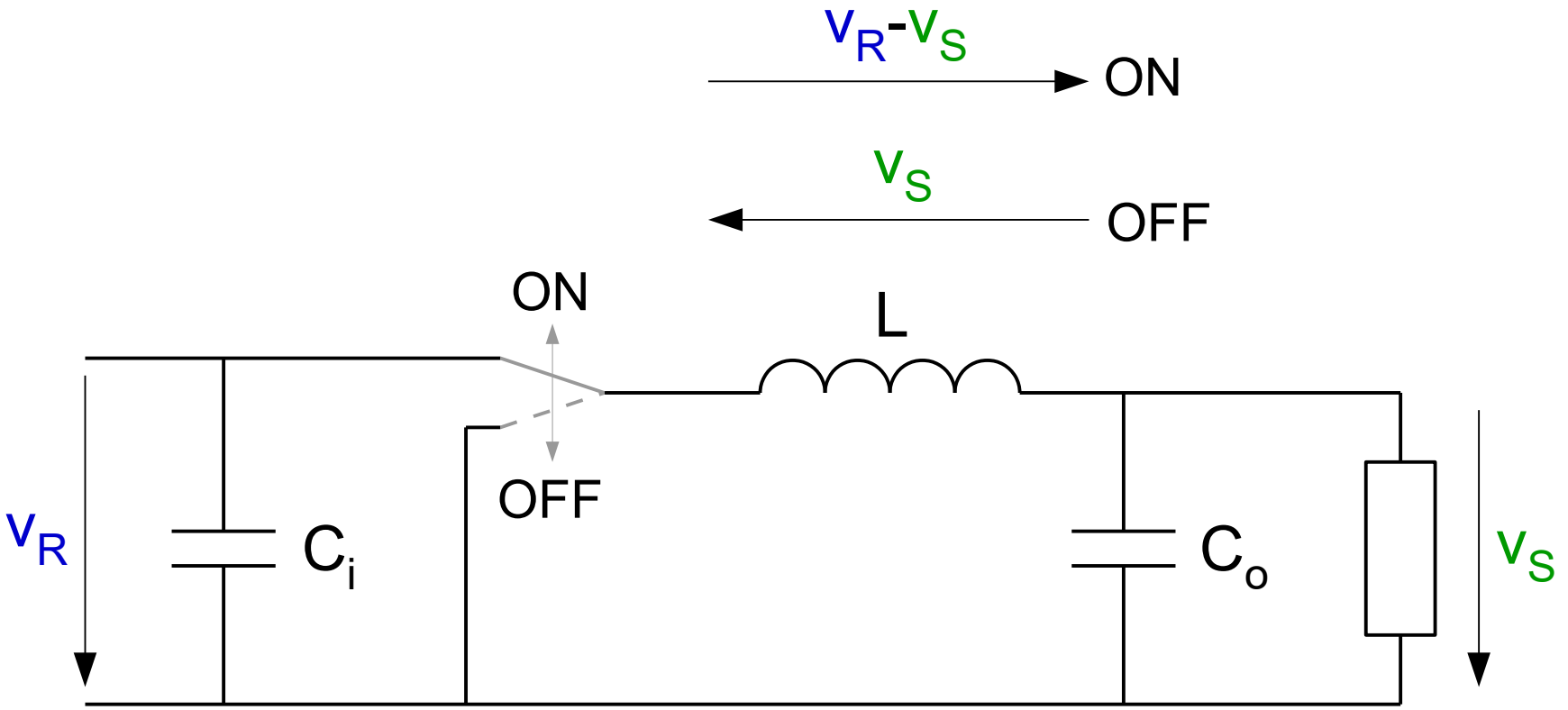
Ideal în curent continuu avem pierdere de putere

$$p_{ERS} = (v_R - v_S)\, i_S$$

Pentru a nu disipa putere avem nevoie de

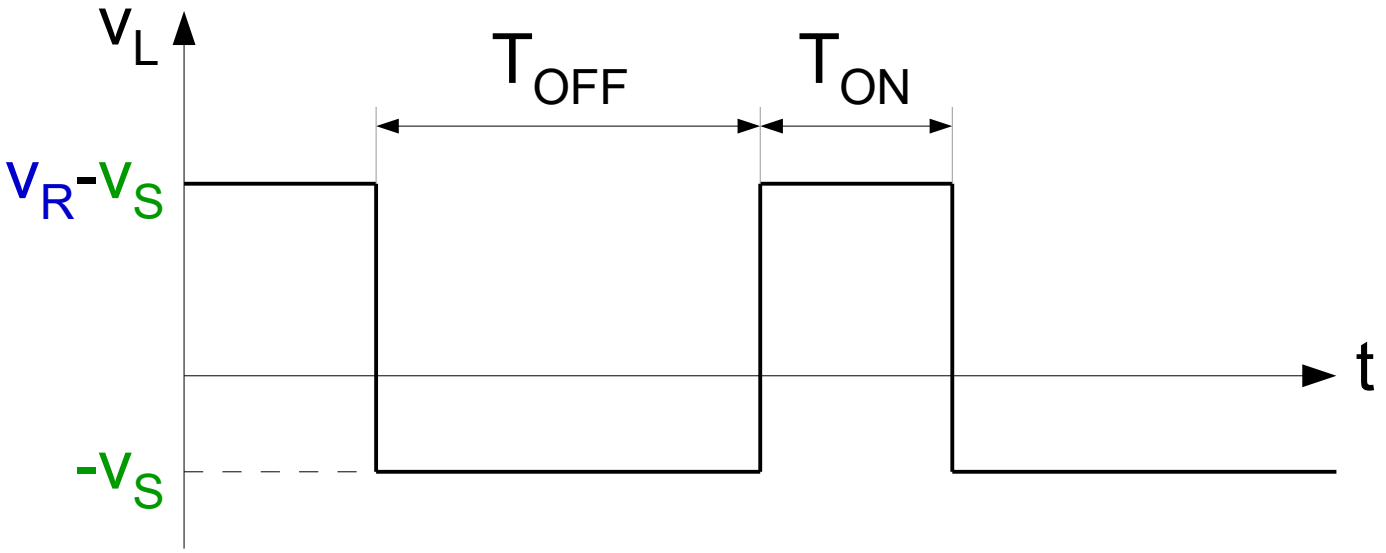
- elemente reactive
- regim alternativ

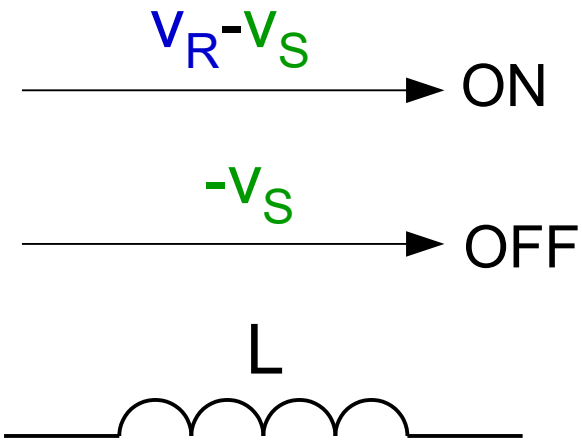
Stabilizator în comutație (2)

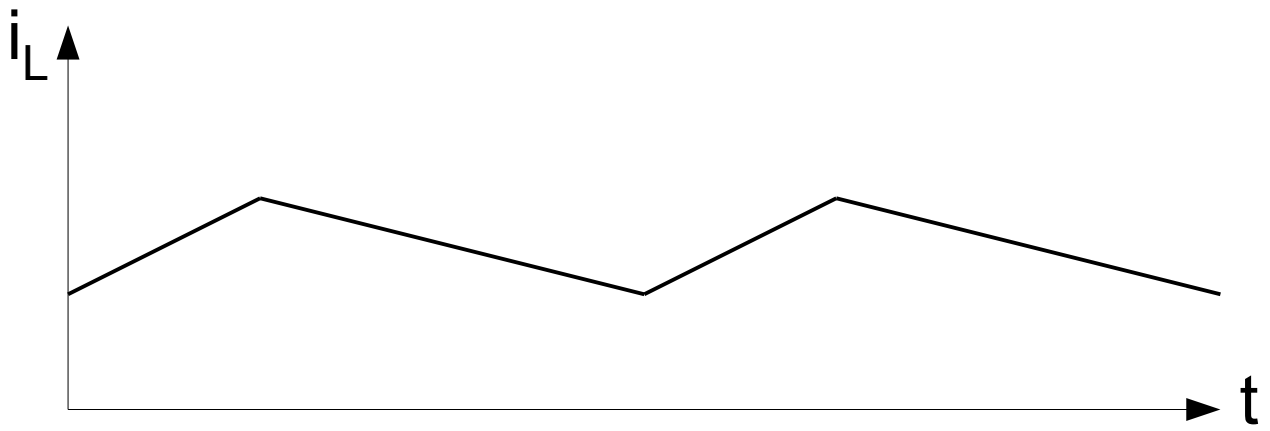


Buck Converter

Stabilizator în comutație (3)

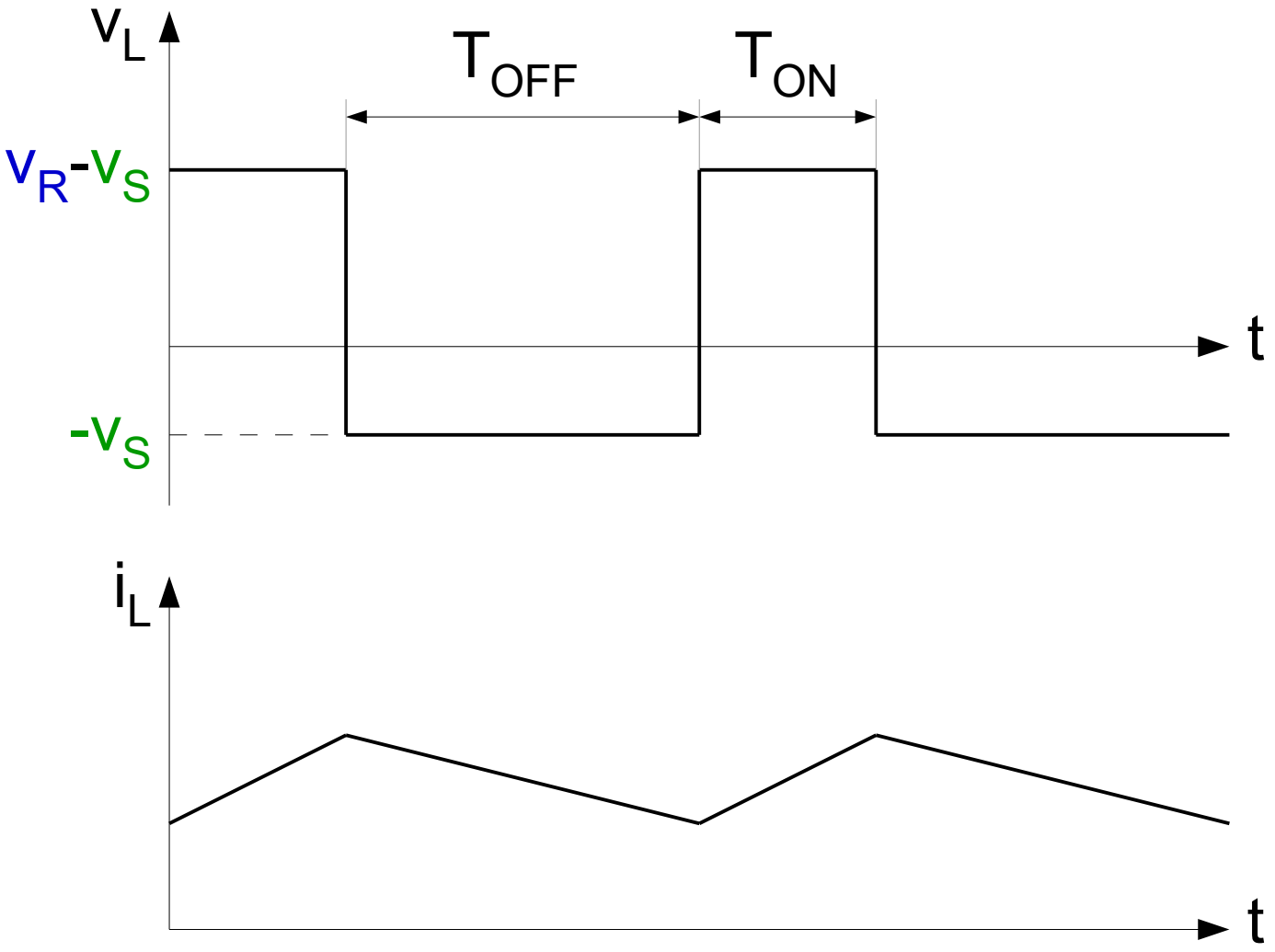






$$v_L = L \frac{d\, i_L}{d\, t}$$

Stabilizator în comutație (4)



factor de umplere (*duty cycle*):

$$D = \frac{T_{ON}}{T_{ON} + T_{OFF}}$$

$$\frac{v_S}{v_R - v_S} = \frac{T_{ON}}{T_{OFF}}$$

$$\boxed{\frac{v_S}{v_R} = D}$$